
TENSORIZED KNOWLEDGE UNITS

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

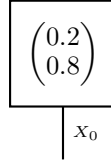
October 6, 2026

ABSTRACT

We investigate, whether tensor networks could serve as a knowledge representation format for the project Alexandria [Schuhmann et al. \(2025\)](#). To represent uncertainties and relations between claims we apply the formalism discussed in [Goessmann et al. \(2026\)](#).

1 Incorporating uncertainties

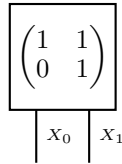
Climate change is happening with 80% probability.



2 Modelling relations between claims

When the polar ice is melting, then climate change is happening.

- X_0 : Polar ice is melting
- X_1 : Climate change is happening



3 Inference

Assuming that the polar ice is melting, what is the probability of climate change, according to the paper?

This would be a contraction with $\epsilon_1 [X_0]$

$$\begin{array}{c} \boxed{\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}} \\ \begin{array}{c} | \\ \epsilon_1 \end{array} \end{array} \begin{array}{c} x_0 \quad x_1 \end{array} = \begin{array}{c} \boxed{\begin{pmatrix} 0 \\ 1 \end{pmatrix}} \\ | \\ x_1 \end{array}$$

4 Similarity by contractions

The extracted tensorized Knowledge Units can be contracted to observe the similarity between their claims. For example, two knowledge units with contradicting claim have vanishing contractions.

References

- Alex Goessmann, Janina Schütte, Maximilian Fröhlich, and Martin Eigel. A tensor network formalism for neuro-symbolic AI, January 2026. URL <http://arxiv.org/abs/2601.15442>. arXiv:2601.15442 [cs].
- Christoph Schuhmann, Gollam Rabby, Ameya Prabhu, Tawsif Ahmed, Andreas Hochlehnert, Huu Nguyen, Nick Akinci, Ludwig Schmidt, Robert Kaczmarczyk, Sören Auer, Jenia Jitsev, and Matthias Bethge. Project Alexandria: Towards Freeing Scientific Knowledge from Copyright Burdens via LLMs, April 2025. URL <http://arxiv.org/abs/2502.19413>. arXiv:2502.19413 [cs.LG].